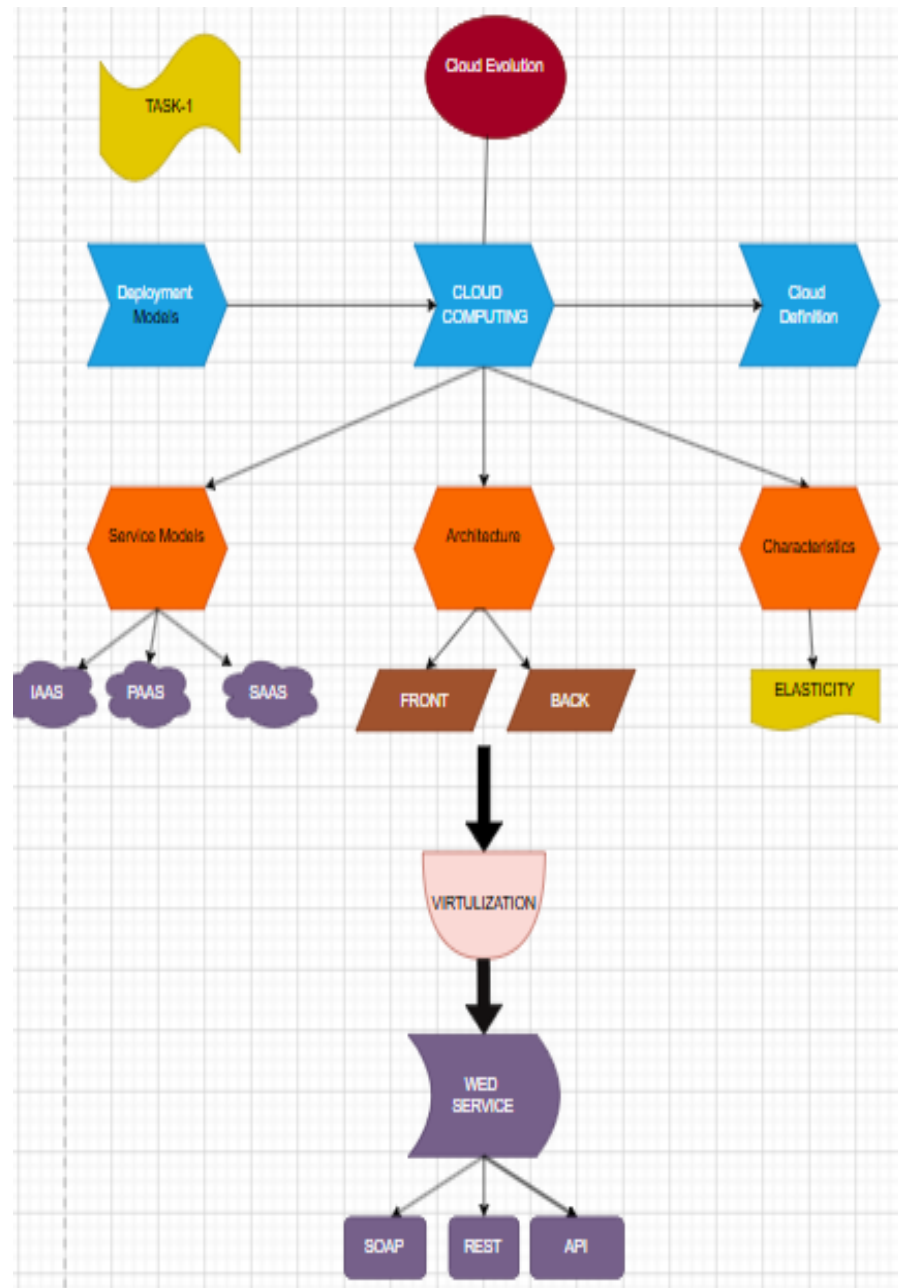


M.GUNA SHEKAR

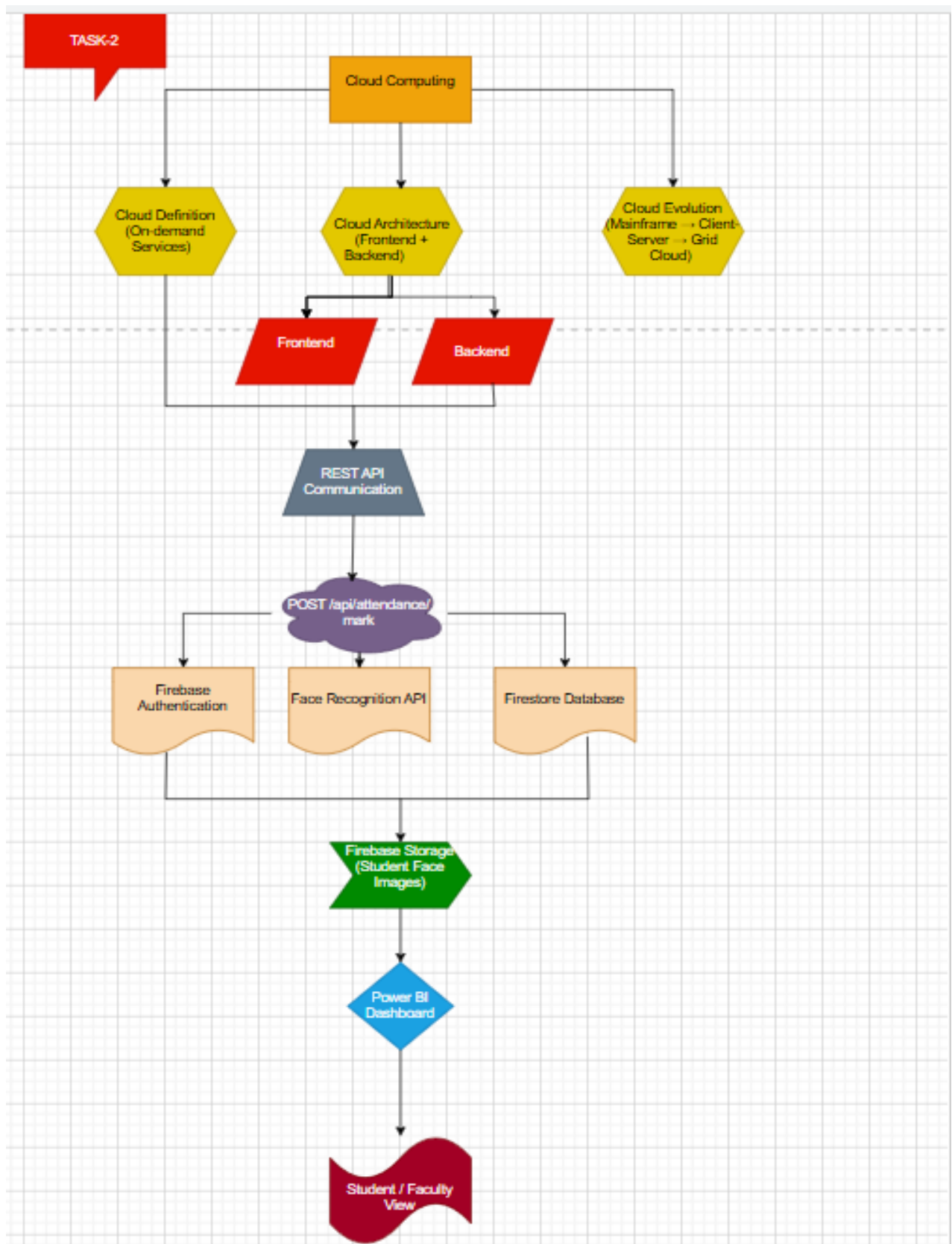
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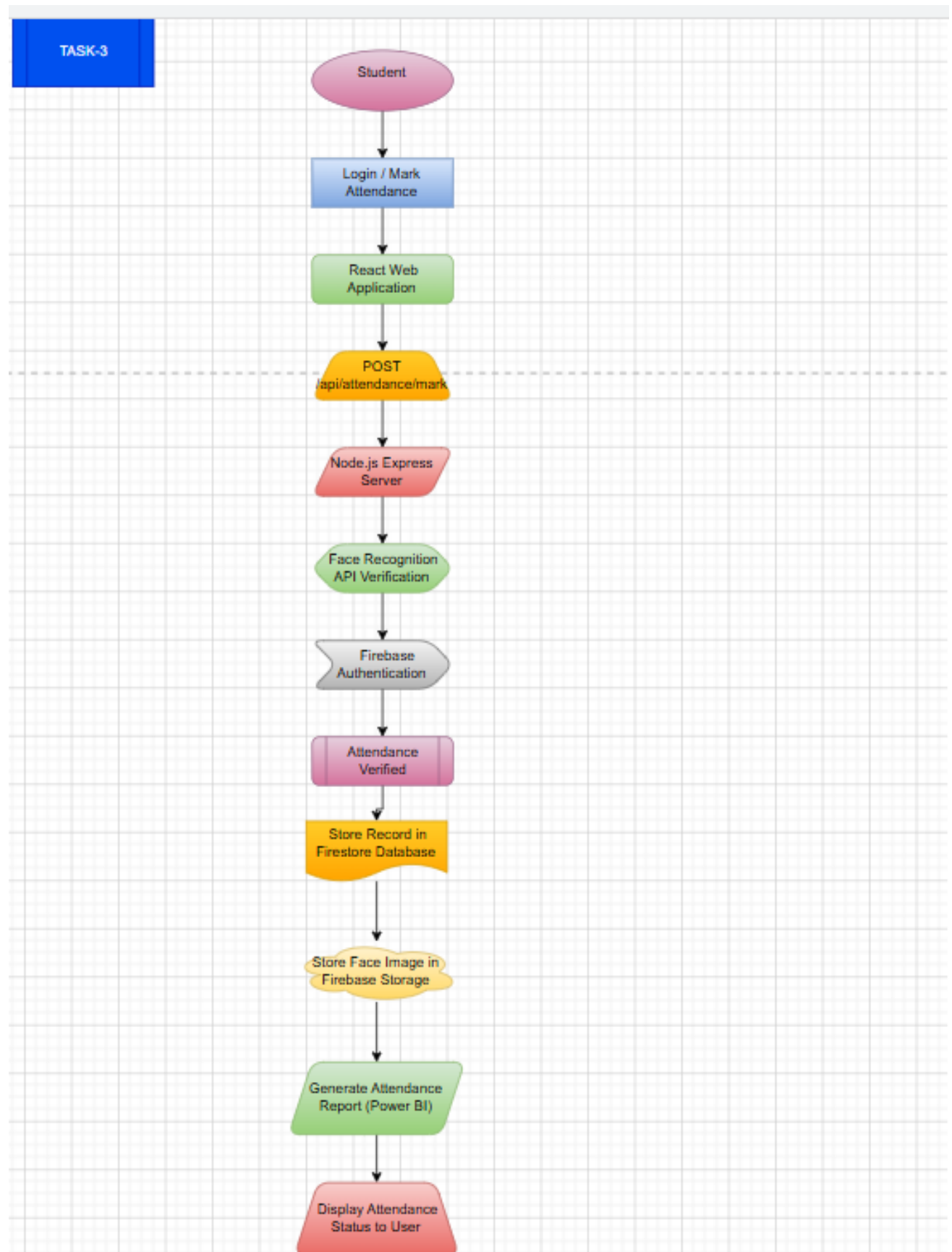
TASK 1 - DIGITAL CLOUD CONCEPT MAP



TASK 2 - SERVICE MODEL MATRIX:-



TASK-3: CAPSTONE CLOUD FLOW MAPPING



Task 4 – Individual Reflection

1. What is the strongest cloud concept represented in your capstone project?

The strongest cloud concept represented in my capstone project is **cloud-based resource sharing**. Cloud computing enables centralized storage, secure access, and real-time synchronization of attendance records from any location. This improves accessibility, collaboration, and overall system reliability while reducing the need for local infrastructure.

2. Where will elasticity or scalability become important in your capstone product?

Elasticity and scalability become important during peak attendance periods, such as the beginning of classes or examination sessions, when many students access the system simultaneously. Cloud auto-scaling allows the application to allocate additional computing resources automatically, ensuring smooth performance and minimizing response time during high-demand periods.

3. Which API or web-service idea is most relevant to your project and why?

REST APIs are the most relevant web-service technology for my project because they enable efficient communication between the React frontend, Node.js backend, Firebase Authentication service, Firestore database, and Face Recognition API. REST APIs provide secure, lightweight, and standardized data exchange, making the system easy to integrate, maintain, and scale.

4. What did you personally understand better after preparing this concept map and matrix?

After preparing the concept map and service matrices, I gained a better understanding of how cloud computing services work together in a real-world application. I learned the differences between IaaS, PaaS, SaaS, and specialized cloud services, as well as the importance of APIs, cloud storage, security, scalability, and virtualization in designing and deploying a complete cloud-based system.